→ Cross-Validation does not replace the final test set.

28-09-2026 (Monday) — Unsupervised learning

Unsupervised learning:- Type of machine learning model, in which a model learn patterns, relationships, and structures from data that has no predefined target / output labels.

Eg:- Customer Age, Annual Income, Total Purchases.

working flow:-

Collected unlabeled data → Process the data

Algo, identifies similarities, differences, (&) hidden structures ← Apply unsupervised learning algorithm

↳ Interpret the discovered patterns (&) use them for business problems.

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① What is Clustering ?- Clustering is an unsupervised learning technique that groups similar datapoints together without predefined labels.

Eg:- Group website visitors based on their browsing behaviour.
- → Casual browser
- → Engaged evaluators.
- → Lapsed users.

→ Why product teams use Clustering :-

not assumptions

1. Segmentation - Group customers based on behaviour,
2. Personalization - Customize recommendations, offers, pricing, messaging (or) on boarding to each group's interests (or) behaviour.
3. Anomaly Detection - Identify data points, that don't fit any cluster.
   ↳ very much similar to "outliers"

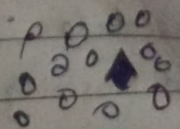
K-mean algorithm :- It sorts data into 'K' groups by (common clustering method) repetedly assigning each point to its nearest center, then moving their centers to the average of its group - until the groups stop changing.

K-mean divides data into a predefined number of clusters, represented by K.

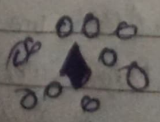
→ 3 main steps :

1. Pick "K" initial cluster centers (Centroid).
2. Assign each data point to its nearest centroid.
3. Update each centroid, using average of its assigned points, the repeat until the cluster stop changing.

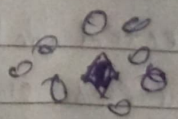
K-means looks like



Casual Browsers



Engaged Evaluators.



Lapsed users

Euclidean distance = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

**Cluster = $\arg\min_k$ distance (visitor, center k)**

This means, assigning each visitor to the nearest cluster center.

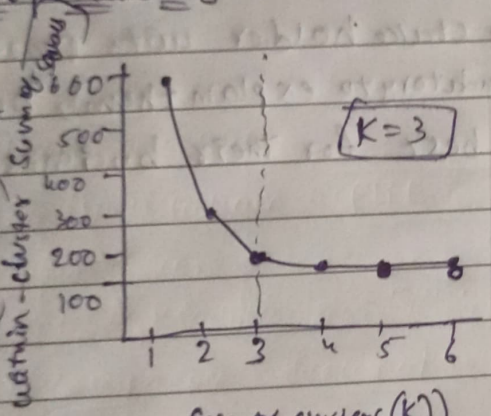
→ How to choose K ?

① Elbow method:- Find the point where adding more clusters stops significantly reducing within-cluster variation.

② Business context:- Choose groups that stacke holders can understand and act on.

③ Interpretability:- More clusters are not automatically better.

Elbow method:




WCSS ⇒ measure, how far Data points are from their assigned cluster centroids.

→ Lower WCSS, mean data points are closer to their cluster centers.

**Inertia** - measure how much close data points are assigned cluster centers (centroids)

$$\text{Inertia} = \sum_{i=1}^{n} \| x_i - \mu_{c_i} \|^2$$

$x_i$ = data point, $\mu_{c_i}$ = centroid of the cluster assigned to that point.

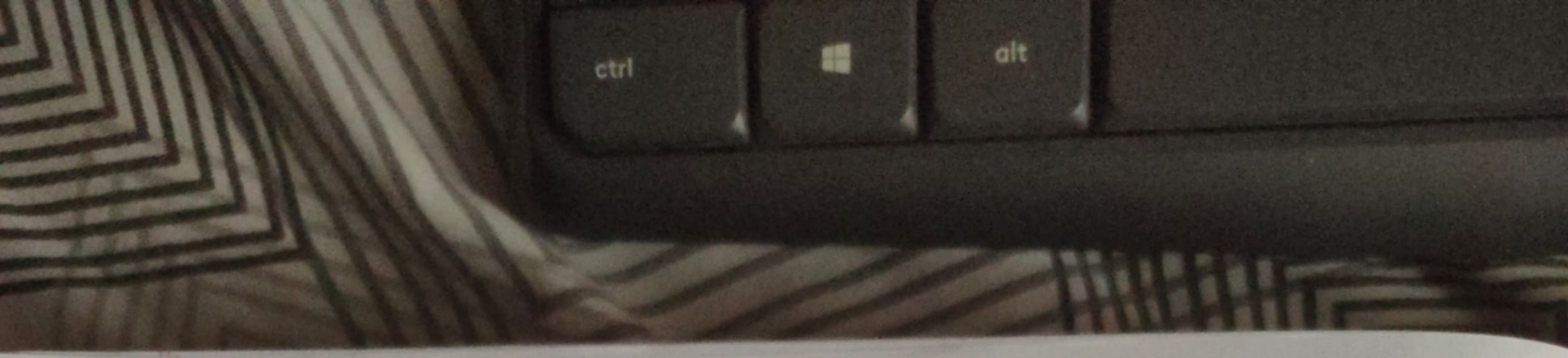
Lower Inertia ⇒ Data points are closer to their cluster centers.

High Inertia ⇒ " " " further from their cluster centers.

Why not choose highest K ?

As K increases, WCSS (I) Inertia, decreases, because each cluster contain fewer points.

But, Creating too many clusters make segmentation complicated (&) less useful for business decisions.



The Elbow method balances :- (1) Low within - cluster variation.
(2) manageble no. of meaningful clusters.

Stack Holder VS elbow method for choosing K :-

Stack Holder are Ultimate decision makers, the Elbow method is a technical guide.

→ If stacke holders think of K=7, elbow method give K=3, Stacke holders revieco & analyse, come up with new 'K' value, ;
→ Only we can convience stacke holder with proof from elbow method (mandatory to explain them), but they decide what suits best for their business.